

Unified Modeling Language (UML)

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Definition

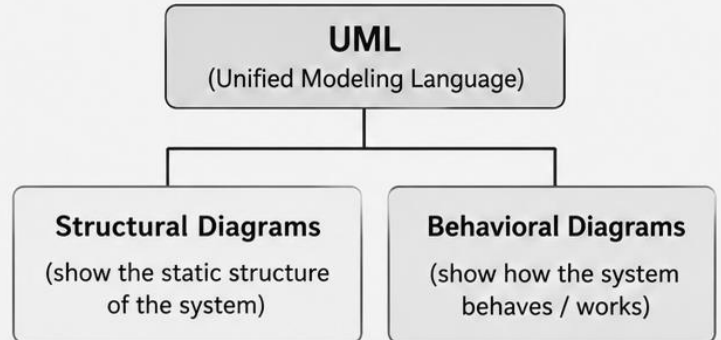
UML (Unified Modeling Language) is a standardized visual modeling language used to specify, visualize, construct and document the artifacts of a software system. It helps in understanding, designing and communicating the structure and behavior of the system in a clear and consistent way.



In simple words:

UML is a set of diagrams and notations used to model a software system.

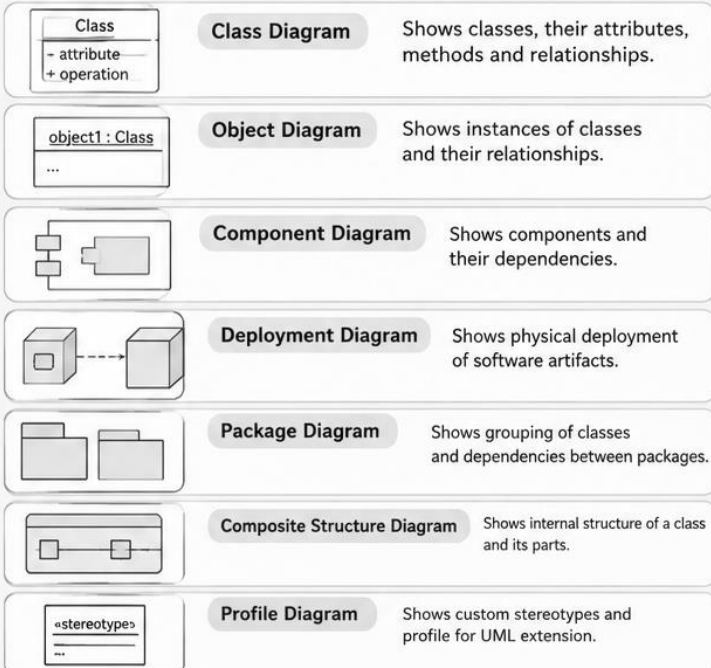
Structure of UML



Types of UML Diagrams

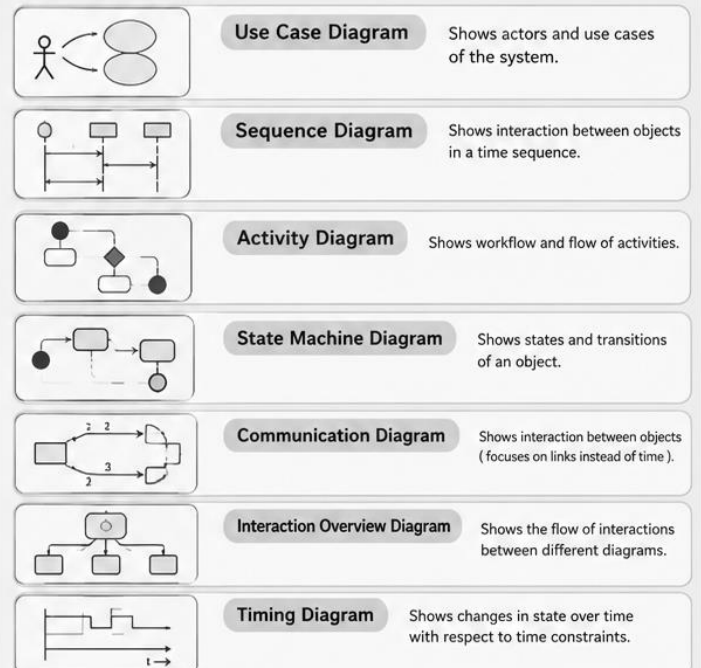
1. Structural Diagrams

These diagrams show the static structure of the system (i.e., what exists).



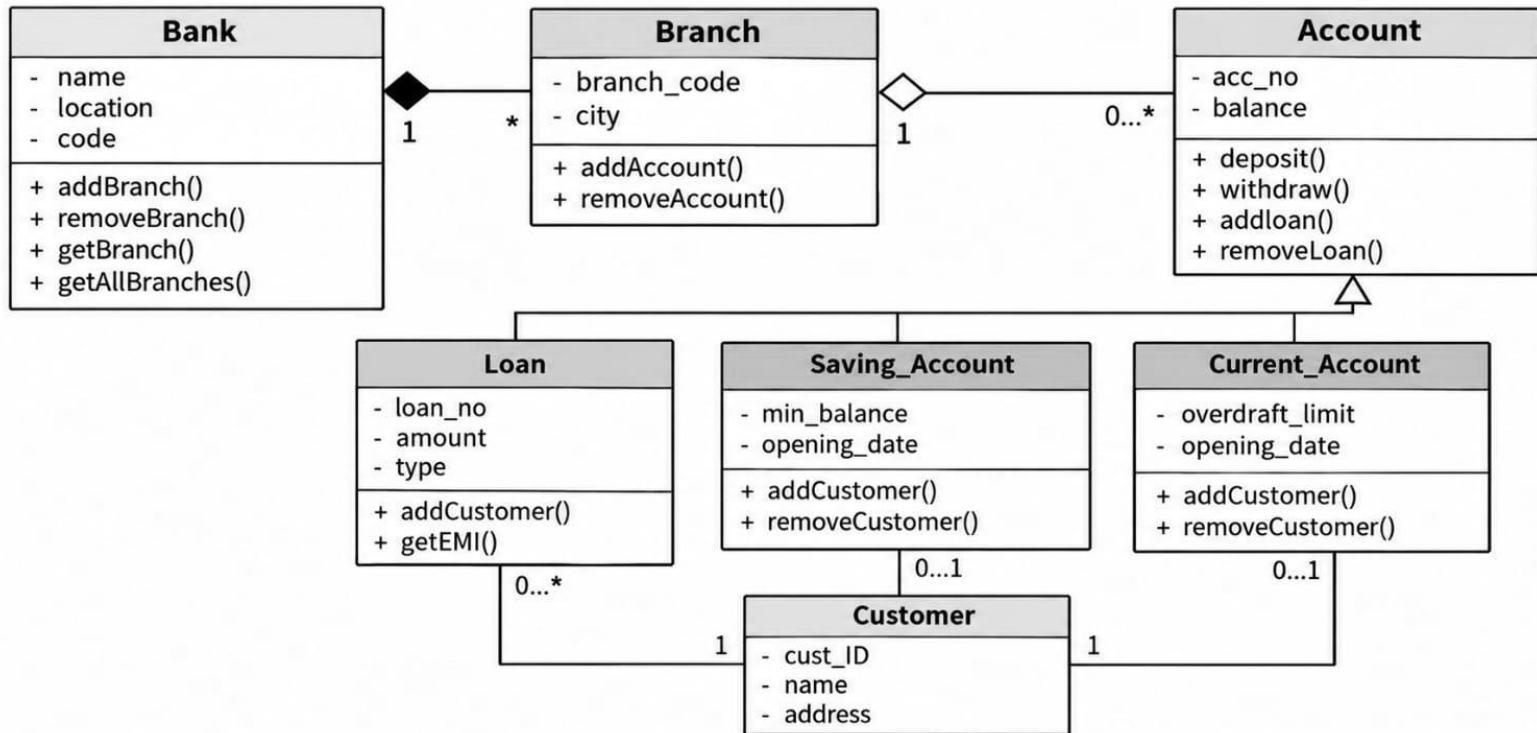
2. Behavioral Diagrams

These diagrams show how the system behaves (i.e., what happens).



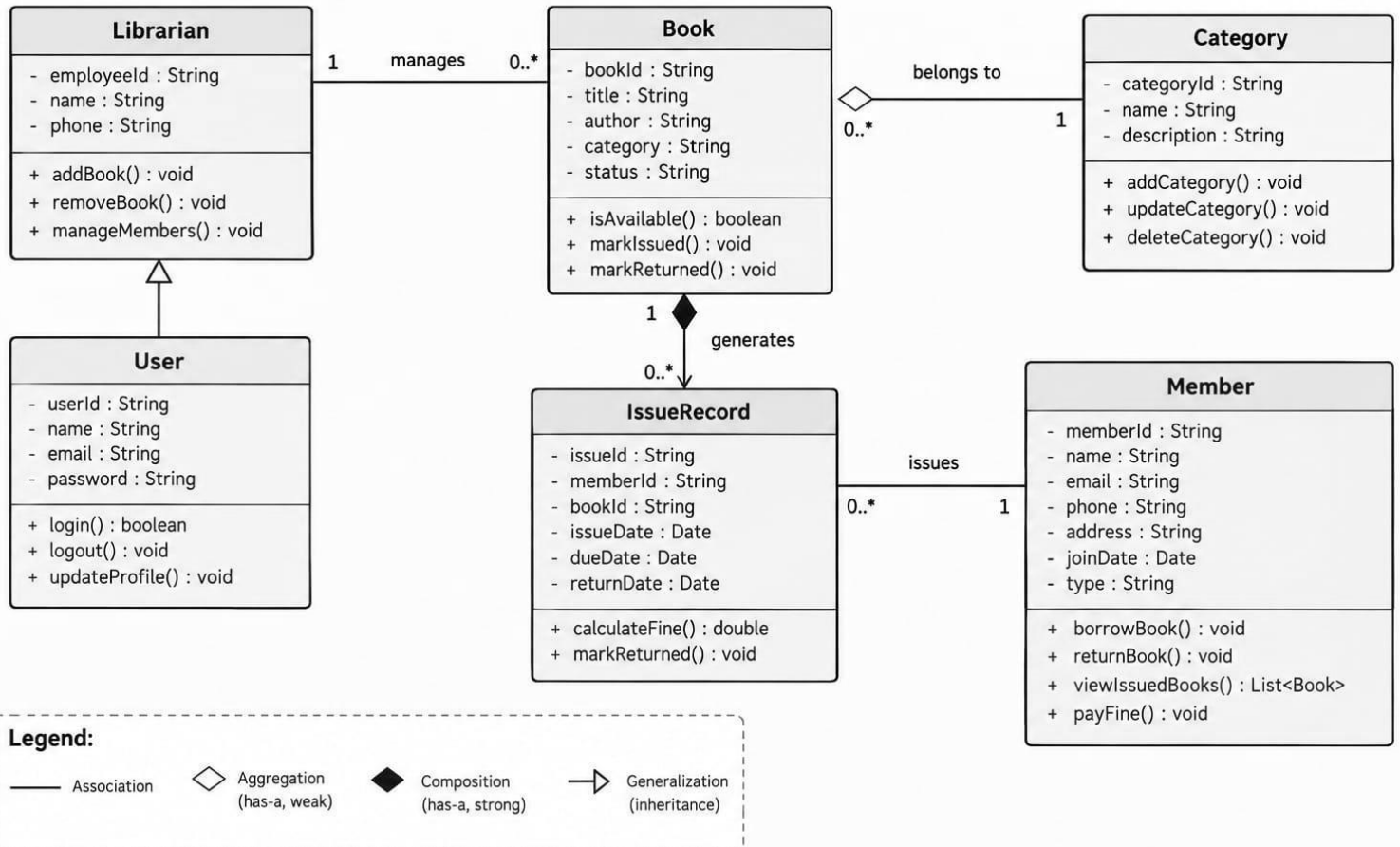
Class Diagrams

Banking System



Library Management System

Library Management System – Class Diagram



OOPS Concepts

....(Explain with definition and Symbols)

Class

Object

Encapsulation

Abstraction

Inheritance

Polymorphism

Association

Aggregation

Composition